

# A constrained eigenvalue problem

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**A Constrained  
Eigenvalue  
Problem**

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1 INTRODUCTION In this paper we consider the following mathematical and computational problem. Given the quantities

A:  $(n + m)$ -by- $(n + m)$  matrix, symmetric,  $n > 0$

N:  $(n + m)$ -by- $m$  matrix with full rank

$\mathbf{t}$  : vector of dimension  $m$  with  $\|(N^T)^+\mathbf{t}\| < 1$

Determine an  $\mathbf{x}$  such that

$$(1) \quad \mathbf{x}^T \mathbf{A} \mathbf{x} = \min$$

subject to the constraints

$$(2) \quad N^T \mathbf{x} = \mathbf{t}$$

$$(3) \quad \mathbf{x}^T \mathbf{x} = 1.$$

Variants of this problem occur in many applications [1,5,7,8,11]. The problem has been studied previously when  $\mathbf{t} = \mathbf{0}$ , the null vector, (cf. [4,6]).

When  $\mathbf{t} \neq \mathbf{0}$ , then the problem becomes more complicated. We now motivate our assumptions. Suppose N does not have full rank. If  $\mathbf{t}$  is not in the range of  $N^T$  the problem has no solution. If, however, the linear constraints are consistent, we can deflate the system until we get sub-matrix of full rank. In the extreme case, where  $N = \mathbf{0}$  and  $\mathbf{t} = \mathbf{0}$ , the problem reduces to the ordinary eigenvalue problem with an eigenvector corresponding to the smallest eigenvalue of A as the solution.

Now consider the quantity  $\|(N^T)^+\mathbf{t}\|$ . As  $(N^T)^+\mathbf{t}$  denotes the unique solution of  $N^T \mathbf{x} = \mathbf{t}$  of minimal norm, we can make the following distinctions: when  $\|(N^T)^+\mathbf{t}\| > 1$  there is no solution. In the case of  $\|(N^T)^+\mathbf{t}\| = 1$  the unique solution is given by  $\mathbf{x} = (N^T)^+\mathbf{t}$ . Therefore, the condition  $\|(N^T)^+\mathbf{t}\| < 1$  is the only interesting case.

Thus when  $\|(N^T)^+\mathbf{t}\| < 1$ , we can always find an  $\mathbf{x}$  that satisfies both constraints and hence there always exists at least one solution to the problem.

**2 PROBLEM SIMPLIFICATION** In this section we will normalize the problem in order to study its solvability. These considerations are not only of theoretical interest, but they also serve as a basis for the numerical calculations.

**2.1 Elimination of the Linear Constraint** In the domain of the transformation  $N^T$  we can distinguish two fundamental subspaces:

$\mathcal{N}(N^T)$  : nullity of  $N^T$

$\mathcal{N}(N^T)^\perp$ : orthogonal complement.

If the transformation  $N^T$  is being restricted on  $\mathcal{N}(N^T)^\perp$  it acts as a bijection between  $\mathcal{N}(N^T)^\perp$  and  $\mathcal{R}(N^T)$  with the inverse  $(N^T)^\perp$ . Thus the general solution of the first constraint is given by

$$\mathbf{x} = (N^T)^\perp \mathbf{t} + \xi$$

with an arbitrary  $\xi \in \mathcal{N}(N^T)$ . Now using the singular value decomposition of  $N^T$ ,

$$N^T = U \Sigma V^T$$

with

$$U = \left[ \begin{array}{c|c} m & \\ \hline U & \end{array} \right] \quad m$$

$$\Sigma = \left[ \begin{array}{ccc|c} m & n & & \\ \hline \sigma_1 & & & 0 \\ & \ddots & & \\ & & \sigma_m & \end{array} \right] \quad m$$

$$V = \left[ \begin{array}{c|c} m & n \\ \hline V_1 & V_2 \end{array} \right] \quad m+n \quad ,$$

we can write  $\mathbf{x}$  as

$$\mathbf{x} = V \Sigma^\perp U^T \mathbf{t} + V_2 \mathbf{z}$$

with an arbitrary  $\mathbf{z}$ . For the subsequent minimization, we only have to consider the second constraint (3).

For practical reasons, we can use the QR-decomposition of  $N$  instead of the singular value decomposition. To simplify the first constraint (2), we write

$$(4) \quad P^T N = \begin{bmatrix} R \\ 0 \end{bmatrix}$$

where  $P$  denotes an orthogonal matrix, and  $R$  is a  $m$ -by- $m$  upper triangular matrix. Now the problem can be formulated as

$$\mathbf{x}^T A \mathbf{x} = \mathbf{x}^T P P^T A P P^T \mathbf{x} = \min$$

$$N^T \mathbf{x} = \begin{bmatrix} R^T & 0 \end{bmatrix} P^T \mathbf{x} = \mathbf{t}$$

$$\mathbf{x}^T \mathbf{x} = \mathbf{x}^T P P^T \mathbf{x} = 1.$$

We now make the definitions,

$$(5) \quad P^T A P =: \begin{bmatrix} m & n \\ B & \Gamma^T \\ \Gamma & C \end{bmatrix} \begin{matrix} m \\ n \end{matrix} ;$$

$$(6) \quad P^T \mathbf{x} =: \begin{bmatrix} \mathbf{y} \\ \mathbf{z} \end{bmatrix} \begin{matrix} m \\ n \end{matrix} .$$

Note that  $C^T = C$ . Now

$$\begin{aligned} \mathbf{x}^T A \mathbf{x} &= \begin{bmatrix} \mathbf{y}^T & \mathbf{z}^T \end{bmatrix} \begin{bmatrix} B & \Gamma^T \\ \Gamma & C \end{bmatrix} \begin{bmatrix} \mathbf{y} \\ \mathbf{z} \end{bmatrix} \\ &= \mathbf{y}^T B \mathbf{y} + \mathbf{y}^T \Gamma^T \mathbf{z} + \mathbf{z}^T \Gamma \mathbf{y} + \mathbf{z}^T C \mathbf{z} \\ &= \mathbf{y}^T B \mathbf{y} + 2\mathbf{z}^T \Gamma \mathbf{y} + \mathbf{z}^T C \mathbf{z}, \\ N^T \mathbf{x} &= \begin{bmatrix} R^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{y} \\ \mathbf{z} \end{bmatrix} = R^T \mathbf{y} = \mathbf{t}, \end{aligned}$$

$$\mathbf{x}^T \mathbf{x} = \begin{bmatrix} \mathbf{y}^T & \mathbf{z}^T \end{bmatrix} \begin{bmatrix} \mathbf{y} \\ \mathbf{z} \end{bmatrix} = \mathbf{y}^T \mathbf{y} + \mathbf{z}^T \mathbf{z} = 1.$$

So we have reduced the constraint (2) to an ordinary linear system with an upper triangular matrix. Now

$$(7) \quad \mathbf{y} = \mathbf{R}^{-T} \mathbf{t}$$

and with the help of the definitions

$$(8) \quad s^2 := 1 - \mathbf{y}^T \mathbf{y} > 0$$

$$(9) \quad \mathbf{b} := -\Gamma \mathbf{y},$$

we get the simplified problem

$$(10) \quad \begin{aligned} \mathbf{z}^T \mathbf{C} \mathbf{z} - 2\mathbf{b}^T \mathbf{z} &= \min \\ \mathbf{z}^T \mathbf{z} &= s^2. \end{aligned}$$

This problem has been extensively studied in the literature (cf. [2, 7, 8, 10]).

*2.2 Stationary Points* In order to calculate the stationary points, we set up the so-called *Lagrange principal function*:

$$(11) \quad \Phi(\mathbf{z}, \lambda) := \mathbf{z}^T \mathbf{C} \mathbf{z} - 2\mathbf{b}^T \mathbf{z} - \lambda(\mathbf{z}^T \mathbf{z} - s^2)$$

Differentiating  $\Phi$  by  $\mathbf{z}$  and  $\lambda$  yields the equations

$$\begin{aligned} 2\mathbf{C} \mathbf{z} - 2\mathbf{b} - 2\lambda \mathbf{z} &= \mathbf{0} \\ \mathbf{z}^T \mathbf{z} - s^2 &= 0 \end{aligned}$$

or normalized

$$(12) \quad \begin{aligned} \mathbf{C} \mathbf{z} &= \lambda \mathbf{z} + \mathbf{b} \\ \mathbf{z}^T \mathbf{z} &= s^2. \end{aligned}$$

Now let us compare the values  $\mathbf{z}^T \mathbf{C} \mathbf{z} - 2\mathbf{b}^T \mathbf{z}$  of different tuples  $(\lambda, \mathbf{z})$ . Following the proof given in [3, 12], a short calculation shows that the smallest  $\lambda$  is needed in order to minimize the value

$\mathbf{z}^T \mathbf{C} \mathbf{z} - 2\mathbf{b}^T \mathbf{z}$ . So in place of the original minimization we can solve the *Lagrange equations*

$$\begin{aligned} \mathbf{C} \mathbf{z} &= \lambda \mathbf{z} + \mathbf{b} \\ \mathbf{z}^T \mathbf{z} &= s^2 \\ \lambda &= \min. \end{aligned} \tag{13}$$

3 SOLVABILITY We will now investigate the solvability of the Lagrange equations (13). Simultaneously this analysis will point out a first method to solve the problem.

3.1 *Explicit Secular Equation* For our discussion, we need the eigenvalue decomposition

$$\mathbf{C} = \mathbf{Q} \mathbf{D} \mathbf{Q}^T \tag{14}$$

where

$$\mathbf{D} = \text{diag}(\delta_1, \dots, \delta_n) \quad \delta_1 \leq \delta_2 \leq \dots \leq \delta_n$$

and

$$\mathbf{Q}^T \mathbf{Q} = \mathbf{I}.$$

Thus the Lagrange equations (13) are transformed as follows:

$$\begin{aligned} \mathbf{Q} \mathbf{D} \mathbf{Q}^T \mathbf{z} &= \lambda \mathbf{Q} \mathbf{Q}^T \mathbf{z} + \mathbf{b} \\ \mathbf{z}^T \mathbf{z} &= \mathbf{z}^T \mathbf{Q} \mathbf{Q}^T \mathbf{z} = s^2. \end{aligned}$$

With the definitions

$$\mathbf{u} := \mathbf{Q}^T \mathbf{z} \tag{15}$$

$$\mathbf{d} := \mathbf{Q}^T \mathbf{b}, \tag{16}$$

this can be simplified to

$$\begin{aligned} \mathbf{D} \mathbf{u} &= \lambda \mathbf{u} + \mathbf{d} \\ \mathbf{u}^T \mathbf{u} &= s^2 \\ \lambda &= \min. \end{aligned} \tag{17}$$



First let us suppose  $\lambda \in \lambda(D)$ . Then there exist diagonal elements  $\delta_i$  with  $\delta_i = \lambda$ . For the ensuing discussion the following index sets turn out to be useful:

$$\begin{aligned} I &:= \{i \mid \delta_i = \lambda\} \\ \bar{I} &:= \{i \mid \delta_i \neq \lambda\} = \{1, \dots, n\} \setminus I. \end{aligned}$$

If there exists a corresponding  $\mathbf{u}$  for such a  $\lambda$ , it must hold that  $\forall i: \delta_i u_i = \lambda u_i + d_i$  with  $\mathbf{u}^T \mathbf{u} = s^2$ . Then for  $i \in I$ , it must be true that  $\lambda u_i = \lambda u_i + d_i$  and this implies  $d_i = 0$ . And for  $i \in \bar{I}$ , it must hold that:

$$u_i = \frac{d_i}{\delta_i - \lambda}.$$

The normalization condition  $\mathbf{u}^T \mathbf{u} = s^2$  can only be satisfied if

$$\sum_{i \in \bar{I}} u_i^2 = \sum_{i \in \bar{I}} \left( \frac{d_i}{\delta_i - \lambda} \right)^2 \leq s^2.$$

As a result we have the following three possibilities:

1. There exists no solution  $\mathbf{u}$  for a given  $\lambda \in \lambda(D)$  if  $\exists i \in I: d_i \neq 0$ , or

$$\sum_{i \in \bar{I}} \left( \frac{d_i}{\delta_i - \lambda} \right)^2 > s^2.$$

2. There exists a unique solution  $\mathbf{u}$  for a given  $\lambda \in \lambda(D)$  if  $\forall i \in I: d_i = 0$  and

$$\sum_{i \in \bar{I}} \left( \frac{d_i}{\delta_i - \lambda} \right)^2 = s^2.$$

Then  $\mathbf{u}$  can be calculated as

$$u_i = \begin{cases} \frac{d_i}{\delta_i - \lambda} & i \in \bar{I} \\ 0 & i \in I \end{cases}.$$

3. There exist several solutions  $\mathbf{u}$  for a given  $\lambda \in \lambda(D)$  if  $\forall i \in \bar{I} : d_i = 0$ , and

$$\sum_{i \in \bar{I}} \left( \frac{d_i}{\delta_i - \lambda} \right)^2 < s^2.$$

The possible values for  $\mathbf{u}$  are given by

$$u_i = \frac{d_i}{\delta_i - \lambda} \quad i \in \bar{I}$$

$$\sum_{i \in I} u_i^2 = s^2 - \sum_{i \in \bar{I}} \left( \frac{d_i}{\delta_i - \lambda} \right)^2 > 0.$$

Thus the set of all solutions  $\mathbf{u}$  constitutes a manifold of dimension  $|I| - 1$  because the  $u_i$  with  $i \in I$  can be chosen arbitrarily on the given hypersphere.

The second case is given by  $\lambda \notin \lambda(D)$ . Then the inverse  $(D - \lambda I)^{-1}$  exists and  $\mathbf{u}$  has the representation

$$(18) \quad \mathbf{u} = (D - \lambda I)^{-1} \mathbf{d}.$$

For  $\mathbf{u}$  to solve the normalization condition  $\mathbf{u}^T \mathbf{u} = s^2$  it must hold that

$$\mathbf{u}^T \mathbf{u} = \mathbf{d}^T (D - \lambda I)^{-2} \mathbf{d} = \sum_{i=1}^n \left( \frac{d_i}{\delta_i - \lambda} \right)^2 = s^2.$$

We define

$$(19) \quad f(\lambda) := \sum_{i=1}^n \left( \frac{d_i}{\delta_i - \lambda} \right)^2 - s^2$$

as the so-called *explicit secular function* (see Figure 1). Thereby the Lagrange equations (13) have a unique solution  $\mathbf{u}$  for a given  $\lambda \notin \lambda(D)$  if and only if the *explicit secular equation*

$$(20) \quad f(\lambda) := \sum_{i=1}^n \left( \frac{d_i}{\delta_i - \lambda} \right)^2 - s^2 = 0$$

is satisfied.

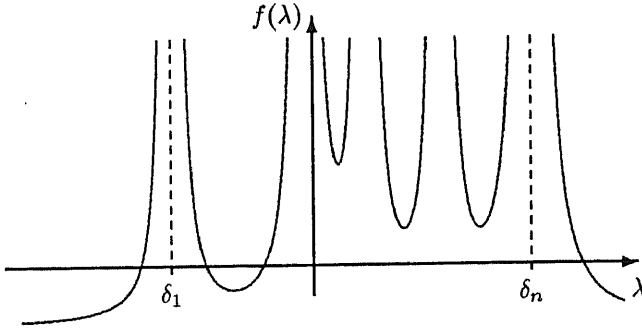


Figure 1: Graph of the secular function

If  $d_i = 0$  for all  $i$ , the secular function (20) degenerates into  $f(\lambda) \equiv -s^2 < 0$  and therefore possesses no solutions. In this case the desired  $\lambda$  lies in the spectrum  $\lambda(D)$ . Thus let  $k$  be the index of the first  $d_i \neq 0$ , i.e.  $d_k \neq 0$  and  $\forall i < k : d_i = 0$ . So we can write the secular function (19) as

$$f(\lambda) = \sum_{i=k}^n \left( \frac{d_i}{\delta_i - \lambda} \right)^2 - s^2$$

with  $\delta_k \neq 0$ . For  $\lambda$  increasing from  $-\infty$  to  $\delta_k$ ,  $f(\lambda)$  increases strictly since the derivative

$$f'(\lambda) = \sum_{i=k}^n \frac{2d_i^2}{(\delta_i - \lambda)^3}$$

is positive for  $-\infty < \lambda < \delta_k$ . From the limits

$$\lim_{\lambda \rightarrow -\infty} f(\lambda) = -s^2$$

$$\lim_{\lambda \rightarrow \delta_k^-} f(\lambda) = +\infty,$$

it immediately follows that for  $\lambda < \delta_k$  there exists exactly one solution.

Then the desired smallest  $\lambda$  can be located either inside or outside the spectrum  $\lambda(D)$ . With the help of the index set  $I := \{i \mid \delta_i = \delta_1\}$  we can distinguish two alternatives:

1. Either it holds that  $\forall i \in I : d_i = 0$ , and

$$\sum_{i \in \bar{I}} \left( \frac{d_i}{\delta_i - \delta_1} \right)^2 \leq s^2.$$

Whence it follows that for  $\lambda = \delta_1$ , there exists a solution of the Lagrange equations (13).  $f(\lambda)$  possesses no more solutions for  $\lambda < \delta_1$  since

$$f(\delta_1) = \sum_{i \in \bar{I}} \left( \frac{d_i}{\delta_i - \delta_1} \right)^2 - s^2 \leq 0.$$

Thus with  $\lambda = \delta_1$  we have found the smallest  $\lambda$ .

2. Or it holds that  $\exists i \in I : d_i \neq 0$ , or

$$\sum_{i \in \bar{I}} \left( \frac{d_i}{\delta_i - \delta_1} \right)^2 > s^2.$$

It follows that  $f(\lambda)$  has a singularity for  $\lambda = \delta_1$ , or else

$$f(\delta_1) = \sum_{i \in \bar{I}} \left( \frac{d_i}{\delta_i - \delta_1} \right)^2 - s^2 > 0.$$

Therefore  $f(\lambda)$  has exactly one solution for  $\lambda < \delta_1$ . This solution represents the desired smallest  $\lambda$ .

Hence, in both alternatives, the smallest  $\lambda$  always satisfies the condition  $\lambda \leq \delta_1$ .

**3.2 Implicit Secular Equation** The above discussion on the location of the smallest  $\lambda$ , that solves the Lagrange equations (13), can be carried out even without the calculation of the eigenvalue decomposition (14) of  $C$ . This is useful, when we want to avoid this factorization numerically.

As indicated before, we know that the desired  $\lambda$  satisfies  $\lambda \leq \delta_1$ . So for  $\lambda = \delta_1$  the following cases can be distinguished:

1. The equation  $Cz = \delta_1 z + \mathbf{b}$  can be inconsistent, i.e.,

$$(C - \delta_1 I)(C - \delta_1 I)^+ \mathbf{b} \neq \mathbf{b}.$$

In this case we have  $\lambda < \delta_1$ , and the *implicit secular equation*

$$(21) \quad f(\lambda) = \mathbf{b}^T (C - \lambda I)^{-2} \mathbf{b} - s^2 = 0$$

must be solved.

2. Now we assume that the equation  $Cz = \delta_1 z + \mathbf{b}$  is consistent. The expression  $(C - \delta_1 I)^+ \mathbf{b}$  represents the solution with smallest norm. If  $\|(C - \delta_1 I)^+ \mathbf{b}\| > s$ , the normalization condition cannot be satisfied, and we have to solve again the secular equation (21).
3. If however  $\|(C - \delta_1 I)^+ \mathbf{b}\| = s$ , we have found the unique solution of the Lagrange equations (13).
4. Finally it can happen that  $\|(C - \delta_1 I)^+ \mathbf{b}\| < s$ . Let  $\xi^{(1)}, \dots, \xi^{(k)}$  denote the  $k$  orthonormal eigenvectors corresponding to the eigenvalue  $\delta_1$ . Since

$$\begin{aligned} (C - \delta_1 I)^+ \mathbf{b} &\perp \mathcal{N}(C - \delta_1 I), \\ \xi^{(i)} &\in \mathcal{N}(C - \delta_1 I), \end{aligned}$$

every vector

$$\mathbf{z} = (C - \delta_1 I)^+ \mathbf{b} + c_1 \xi^{(1)} + \dots + c_k \xi^{(k)}$$

with

$$c_1^2 + \dots + c_k^2 = s^2 - \|(C - \delta_1 I)^+ \mathbf{b}\|^2$$

solves the Lagrange equations (13). Therefore the set of solutions constitutes a manifold of dimension  $k - 1$ .

**3.3 Condition of the Secular Equation** The calculation of the smallest zero  $\lambda$  of the secular equation (20, 21) is a delicate procedure. Even small errors  $\Delta\lambda$  can result in large deviations  $\Delta\mathbf{x}$

at the solution  $\mathbf{x}$  and  $\Delta \min$  at the minimal value  $\min$ . To illustrate the point we will approximately determine the deviations  $\Delta \mathbf{x}$  and  $\Delta \min$  if  $\Delta \lambda$  is given. We will assume that for small  $\Delta \lambda$  the quantities  $\Delta \mathbf{x}$  and  $\Delta \min$  are essentially linearly dependent of  $\Delta \lambda$ .

Starting with the smallest zero  $\lambda$  of the explicit secular equation (20),  $\mathbf{x}$  and  $\min$  are computed as follows:  $\mathbf{u} := (\mathbf{D} - \lambda \mathbf{I})^{-1} \mathbf{d}$ ,  $\mathbf{z} := \mathbf{Q} \mathbf{u}$ ,

$$\mathbf{x} := \mathbf{P} \begin{bmatrix} \mathbf{y} \\ \mathbf{z} \end{bmatrix}, \quad \text{and}$$

$$\min := \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} \mathbf{y}^T & \mathbf{z}^T \end{bmatrix} \begin{bmatrix} \mathbf{B} & \mathbf{\Gamma}^T \\ \mathbf{\Gamma} & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{y} \\ \mathbf{z} \end{bmatrix}.$$

If instead of the theoretical zero  $\lambda$ , the value  $\lambda + \Delta \lambda$  is used the result is as follows:

$$\begin{aligned} \mathbf{u} + \Delta \mathbf{u} &= (\mathbf{D} - (\lambda + \Delta \lambda) \mathbf{I})^{-1} \mathbf{d} \\ &= (\mathbf{D} - \lambda \mathbf{I})^{-1} \mathbf{d} + (\mathbf{D} - \lambda \mathbf{I})^{-2} \mathbf{d} \Delta \lambda \\ &\quad + O(\Delta \lambda^2). \end{aligned}$$

From this it follows that  $\Delta \mathbf{u} \approx (\mathbf{D} - \lambda \mathbf{I})^{-2} \mathbf{d} \Delta \lambda$ . Now, we have  $\mathbf{z} + \Delta \mathbf{z} = \mathbf{Q}(\mathbf{u} + \Delta \mathbf{u})$ , so  $\Delta \mathbf{z} = \mathbf{Q} \Delta \mathbf{u}$ , and

$$\mathbf{x} + \Delta \mathbf{x} = \mathbf{P} \begin{bmatrix} \mathbf{y} \\ \mathbf{z} + \Delta \mathbf{z} \end{bmatrix}$$

implies

$$\Delta \mathbf{x} = \mathbf{P} \begin{bmatrix} \mathbf{0} \\ \Delta \mathbf{z} \end{bmatrix}.$$

Finally

$$\begin{aligned} \min + \Delta \min &= (\mathbf{x} + \Delta \mathbf{x})^T \mathbf{A} (\mathbf{x} + \Delta \mathbf{x}) \\ &= \mathbf{x}^T \mathbf{A} \mathbf{x} + 2 \mathbf{x}^T \mathbf{A} \Delta \mathbf{x} + \Delta \mathbf{x}^T \mathbf{A} \Delta \mathbf{x} \end{aligned}$$

so that

$$\Delta \min \approx 2\mathbf{x}^T \mathbf{A} \Delta \mathbf{x}.$$

Now considering that

$$\begin{aligned} 2\mathbf{x}^T \mathbf{A} \Delta \mathbf{x} &= 2\mathbf{x}^T \mathbf{P} \mathbf{P}^T \mathbf{A} \mathbf{P} \mathbf{P}^T \Delta \mathbf{x} \\ &= 2 \begin{bmatrix} \mathbf{y}^T & \mathbf{z}^T \end{bmatrix} \begin{bmatrix} \mathbf{B} & \mathbf{\Gamma}^T \\ \mathbf{\Gamma} & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{0} \\ \Delta \mathbf{z} \end{bmatrix} \\ &= 2\mathbf{y}^T \mathbf{\Gamma}^T \Delta \mathbf{z} + 2\mathbf{z}^T \mathbf{C} \Delta \mathbf{z} \\ &= 2(\mathbf{z}^T \mathbf{C} - \mathbf{b}^T) \Delta \mathbf{z}, \end{aligned}$$

we get for the deviations  $\Delta \mathbf{x}$  and  $\Delta \min$  the values

$$\begin{aligned} \Delta \mathbf{x} &= \mathbf{P} \begin{bmatrix} 0 \\ \mathbf{Q}(\mathbf{D} - \lambda \mathbf{I})^{-2} \mathbf{d} \end{bmatrix} \Delta \lambda \\ (22) \quad &\equiv \kappa(\mathbf{x}) \Delta \lambda, \quad \text{and} \end{aligned}$$

$$\begin{aligned} \Delta \min &= 2(\mathbf{z}^T \mathbf{C} - \mathbf{b}^T) \mathbf{Q}(\mathbf{D} - \lambda \mathbf{I})^{-2} \mathbf{d} \Delta \lambda \\ (23) \quad &\equiv \kappa(\min) \Delta \lambda. \end{aligned}$$

Here the quantities

$$(24) \quad \kappa(\mathbf{x}) := \mathbf{P} \begin{bmatrix} 0 \\ \mathbf{Q}(\mathbf{D} - \lambda \mathbf{I})^{-2} \mathbf{d} \end{bmatrix}$$

$$(25) \quad \kappa(\min) := 2(\mathbf{z}^T \mathbf{C} - \mathbf{b}^T) \mathbf{Q}(\mathbf{D} - \lambda \mathbf{I})^{-2} \mathbf{d}$$

denote the *condition vector* of  $\mathbf{x}$  and the *condition number* of  $\min$ . In the actual computation  $\kappa(\mathbf{x})$  and  $\kappa(\min)$  can be calculated as well, and one can get an estimate of the numerical error.

The value of  $\|\kappa(\mathbf{x})\|$  is bounded as follows:

$$\begin{aligned} \|\kappa(\mathbf{x})\| &= \left\| P \begin{bmatrix} \mathbf{0} \\ Q(D - \lambda I)^{-2} \mathbf{d} \end{bmatrix} \right\| \\ (26) \quad &\leq \|(D - \lambda I)^{-2}\| \|\mathbf{d}\| = \frac{\|\mathbf{d}\|}{(\delta_1 - \lambda)^2}. \end{aligned}$$

If  $\mathbf{d}$  happens to be an eigenvector to the eigenvalue  $\delta_1$  of  $D$ , then we have an equality. Similarly, we can bound the condition number of  $\min$ :

$$\begin{aligned} |\kappa(\min)| &= |2\mathbf{x}^T A P \begin{bmatrix} \mathbf{0} \\ Q(D - \lambda I)^{-2} \mathbf{d} \end{bmatrix}| \\ &< 2\|A\| \|(D - \lambda I)^{-2}\| \|\mathbf{d}\| \\ (27) \quad &= 2\|A\| \frac{\|\mathbf{d}\|}{(\delta_1 - \lambda)^2}. \end{aligned}$$

However, we have a strict inequality now. We can prove this by contradiction. Without loss of generality, let us assume  $P = I$  and  $Q = I$ . That is, we could apply the transformations (4) and (14) beforehand and start right away with a matrix  $A$ , where the trailing  $n$ -by- $n$  submatrix is diagonal, and with an upper triangular matrix  $N$ . Suppose that we are given a problem where the bound on  $|\kappa(\min)|$  is attained. This essentially implies, that

$$|\mathbf{x}^T A \Delta \mathbf{x}| = \|\mathbf{x}\| \|A\| \|\Delta \mathbf{x}\|.$$

This equality can only be satisfied if  $\Delta \mathbf{x}$  is an eigenvector to the largest eigenvalue (in absolute value) of  $A$ . Furthermore,  $\mathbf{x}$  and  $\Delta \mathbf{x}$  must be parallel, i.e.  $\mathbf{x}$  is a multiple of  $\Delta \mathbf{x}$ . As the first  $m$  elements of  $\Delta \mathbf{x}$  are zero, so are the corresponding elements of  $\mathbf{x}$ . (6) implies  $\mathbf{y} = \mathbf{0}$ , and from (9) it follows that  $\mathbf{b} = \mathbf{0}$ . But from (16) we have  $\mathbf{d} = \mathbf{0}$ , so the desired smallest  $\lambda$  of the Lagrange equations (13) lies in the spectrum  $\lambda(C)$ , and we would not be solving the secular equation at all.



Thus it is obvious that we have to face large errors if the smallest zero  $\lambda$  of the secular equation (20, 21) is near the smallest eigenvalue  $\delta_1$  of  $C$ . Since the norm of the matrix  $A$  is normally bigger than 1 an inaccurately determined zero affects the minimal value  $\min$  more than the solution vector  $\mathbf{x}$ .

4 ZERO FINDER Now, we want to calculate the smallest zero of the secular equation (20) with  $d_k \neq 0$ . We will solve it by using an iterative method. Suppose we know an approximation  $\lambda^{(i)}$ . Then we can approximate  $f(\lambda)$  with the replacement function

$$(28) \quad g(\lambda) = \frac{a}{(b - \lambda)^2} - s^2$$

in such a way that

$$\begin{aligned} g(\lambda^{(i)}) &= f(\lambda^{(i)}) \\ g'(\lambda^{(i)}) &= f'(\lambda^{(i)}). \end{aligned}$$

The zero of  $g(\lambda)$  will determine the next approximation  $\lambda^{(i+1)}$ .

A short calculation yields the values

$$\begin{aligned} a &= 4 \frac{(f(\lambda^{(i)}) + s^2)^3}{f'^2(\lambda^{(i)})} \\ b &= \lambda^{(i)} + 2 \frac{f(\lambda^{(i)}) + s^2}{f'(\lambda^{(i)})} \end{aligned}$$

and for the zero  $\lambda^{(i+1)} = b - \sqrt{a}/s$  of the replacement function  $g(\lambda)$  we get

$$\lambda^{(i+1)} = \lambda^{(i)} - 2 \frac{f(\lambda^{(i)}) + s^2}{f'(\lambda^{(i)})} \left( \frac{\sqrt{f(\lambda^{(i)}) + s^2}}{s} - 1 \right).$$

It can be shown that this iteration process will yield a strictly decreasing sequence of approximations  $\lambda^{(i)}$ . The reader is referred to [9,12].

**4.1 Initial Value** Now, in order to start the iteration we need to construct an initial value. For a first guess the reduced secular equation

$$\left( \frac{d_k}{\delta_k - \lambda} \right)^2 - s^2 = 0$$

is useful. This leads to the initial value

$$\lambda^{(0)} = \delta_k - \frac{|d_k|}{s}.$$

For this  $\lambda^{(0)}$  it is obvious that  $\lambda^{(0)} < \delta_k$  with  $f(\lambda^{(0)}) \geq 0$ .

**4.2 Stopping Criterion** In theory, the iteration process yields a strictly decreasing sequence of approximations  $\lambda^{(i)}$ , but this property does not persist in finite arithmetic. It is therefore reasonable to terminate the iteration when the strict monotonicity is lost, namely if  $\lambda^{(i+1)} \geq \lambda^{(i)}$ . This method has the advantage that it is machine-independent and that it does not need any knowledge of machine accuracy.

**4.3 Implicit Secular Equation** If we do not want to compute the eigenvalue decomposition (14) of  $C$ , we have to consider the evaluation of the *implicit secular function*

$$(29) \quad f(\lambda) = \mathbf{b}^T (C - \lambda I)^{-2} \mathbf{b} - s^2$$

and its derivative

$$f'(\lambda) = 2\mathbf{b}^T (C - \lambda I)^{-3} \mathbf{b}.$$

With the definitions

$$\mathbf{u} := (C - \lambda I)^{-1} \mathbf{b}$$

$$\mathbf{u}' := (C - \lambda I)^{-1} \mathbf{u},$$

these values can be expressed as

$$f(\lambda) = \mathbf{u}^T \mathbf{u} - s^2$$

$$f'(\lambda) = 2\mathbf{u}^T \mathbf{u}'.$$

Therefore each iteration step requires the solution of two linear systems with the matrix  $C - \lambda I$ .

Again with the explicit secular equation, the quantity

$$\lambda^{(0)} = \delta_k - \frac{|d_k|}{s}$$

yields an initial value with  $f(\lambda^{(0)}) \geq 0$  and  $\lambda^{(0)} < \delta_k$ . If the column vector  $\mathbf{q}_k$  of  $Q$  were known, we could compute the value  $d_k$ , since (16) implies  $d_k = \mathbf{q}_k^T \mathbf{b}$ . If, however, we have an eigenvector  $\xi$  with  $\|\xi\| = 1$  corresponding to the smallest eigenvalue  $\delta_1$  of  $C$ , it is straightforward to use the quantity

$$\lambda^{(0)} := \delta_1 - \frac{|\xi^T \mathbf{b}|}{s}$$

as an initial value. Then it can be shown [12] that  $f(\lambda^{(0)}) \geq 0$  and  $\lambda^{(0)} < \delta_1$  holds when  $\xi^T \mathbf{b} \neq 0$ .

**5 QUADRATIC EIGENVALUE PROBLEM** The two previously mentioned methods have the property that they reduce the problem to finding the solution of a one-dimensional secular equation. Considering the problem from another point of view the Lagrange equations (13) can be reduced to a quadratic eigenvalue problem. For the derivation let us assume  $\lambda \notin \lambda(C)$ . In this case  $\mathbf{z}$  can be written as

$$\mathbf{z} = (C - \lambda I)^{-1} \mathbf{b}.$$

Taking into account the normalization condition for  $\mathbf{z}$  we get the secular function

$$f(\lambda) = \mathbf{b}^T (C - \lambda I)^{-2} \mathbf{b} - s^2,$$

of which the zeroes are to be computed. The task looks different if we make the definition

$$\gamma := (C - \lambda I)^{-2} \mathbf{b}$$

so that

$$(C - \lambda I)^2 \gamma = \mathbf{b}.$$

Instead of the secular equations, we have to solve the system

$$(30) \quad \mathbf{b}^T \gamma - s^2 = 0$$

$$(31) \quad (\mathbf{C} - \lambda \mathbf{I})^2 \gamma = \mathbf{b}.$$

The first condition (30) can also be formulated as

$$1 = \frac{1}{s^2} \mathbf{b}^T \gamma.$$

Using this factor 1 as a coefficient of  $\mathbf{b}$  in (31) we get the *quadratic eigenvalue problem*

$$(32) \quad (\mathbf{C} - \lambda \mathbf{I})^2 \gamma = \frac{1}{s^2} \mathbf{b} \mathbf{b}^T \gamma.$$

Note that the restriction  $\lambda \notin \lambda(\mathbf{C})$  is no longer necessary. Of course, we must face the fact that the set of solutions for  $\lambda$  has been extended by these manipulations, for two equations cannot be formulated as a single one without consequences. Subsequently we will compare the solutions of the quadratic eigenvalue problem (32) with those of the Lagrange equations (13).

**5.1 Solvability** We show the following. Assume  $\lambda$  and  $\mathbf{z}$  fulfil the Lagrange equations

$$\begin{aligned} \mathbf{C}\mathbf{z} &= \lambda\mathbf{z} + \mathbf{b} \\ \mathbf{z}^T \mathbf{z} &= s^2. \end{aligned}$$

**THEOREM 5.1.** The quadratic eigenvalue problem

$$(\mathbf{C} - \lambda \mathbf{I})^2 \gamma = \frac{1}{s^2} \mathbf{b} \mathbf{b}^T \gamma$$

has a solution for this  $\lambda$ .

**PROOF.** In our proof, we have to distinguish whether  $\lambda$  lies in the spectrum  $\lambda(\mathbf{C})$  or not.

**Case 1:**  $\lambda \in \lambda(\mathbf{C})$ . Let  $\gamma$  be an eigenvector of  $\mathbf{C}$  to the eigenvalue  $\lambda$ . Then  $\mathbf{b} = (\mathbf{C} - \lambda \mathbf{I})\mathbf{z}$  implies  $\mathbf{b}^T \gamma = \mathbf{z}^T (\mathbf{C} - \lambda \mathbf{I})\gamma = 0$ .

From this it follows that

$$(C - \lambda I)^2 \gamma = 0$$

$$\frac{1}{s^2} \mathbf{b} \mathbf{b}^T \gamma = 0,$$

and hence  $\gamma$  satisfies the eigenvalue equation.

**Case 2:**  $\lambda \notin \lambda(C)$ . With the definition

$$\gamma := (C - \lambda I)^{-1} \mathbf{z}$$

it follows that

$$(C - \lambda I)^2 \gamma = (C - \lambda I) \mathbf{z} = \mathbf{b}$$

$$\frac{1}{s^2} \mathbf{b} \mathbf{b}^T \gamma = \frac{1}{s^2} \mathbf{b} \mathbf{b}^T (C - \lambda I)^{-1} \mathbf{z}$$

$$= \frac{1}{s^2} \mathbf{b} \mathbf{z}^T \mathbf{z} = \mathbf{b},$$

and again  $\gamma$  satisfies the eigenvalue equation.

Therefore, we can construct a solution  $\gamma$  of the quadratic eigenvalue problem (32) in both cases.  $\square$

Conversely, we can assume that  $\lambda$  and  $\gamma$  fulfil the quadratic eigenvalue equation

$$(C - \lambda I)^2 \gamma = \frac{1}{s^2} \mathbf{b} \mathbf{b}^T \gamma, \quad \lambda \notin \lambda(C).$$

**THEOREM 5.2.**  $\lambda$  and  $\mathbf{z} := (C - \lambda I)^{-1} \mathbf{b}$  fulfil the Lagrange equations

$$C\mathbf{z} = \lambda \mathbf{z} + \mathbf{b}$$

$$\mathbf{z}^T \mathbf{z} = s^2.$$

**PROOF.** Obviously the first equation is satisfied. Multiplying the quadratic eigenvalue equation (32) by  $(C - \lambda I)^{-2}$  we get

$$\gamma = \frac{1}{s^2} (\mathbf{b}^T \gamma) (C - \lambda I)^{-2} \mathbf{b} \neq 0.$$

This implies  $\mathbf{b}^T \gamma \neq 0$  and

$$(\mathbf{C} - \lambda \mathbf{I})^{-2} \mathbf{b} = \frac{s^2}{\mathbf{b}^T \gamma} \gamma.$$

The square of the norm of  $\mathbf{z}$  becomes:

$$\mathbf{z}^T \mathbf{z} = \mathbf{b}^T (\mathbf{C} - \lambda \mathbf{I})^{-2} \mathbf{b} = \mathbf{b}^T \frac{s^2}{\mathbf{b}^T \gamma} \gamma = s^2 \frac{\mathbf{b}^T \gamma}{\mathbf{b}^T \gamma} = s^2,$$

with which the second equation is satisfied too.  $\square$

Finally, we assume that  $\lambda$  and  $\gamma$  fulfil the quadratic eigenvalue equation

$$(\mathbf{C} - \lambda \mathbf{I})^2 \gamma = \frac{1}{s^2} \mathbf{b} \mathbf{b}^T \gamma, \quad \lambda \in \lambda(\mathbf{C}).$$

We define  $\mathbf{u} := (\mathbf{C} - \lambda \mathbf{I})^+ \mathbf{b}$ .

**THEOREM 5.3.** For the solvability of the Lagrange equations

$$\begin{aligned} \mathbf{C} \mathbf{z} &= \lambda \mathbf{z} + \mathbf{b} \\ \mathbf{z}^T \mathbf{z} &= s^2 \end{aligned}$$

we can make the following distinction: If the Lagrange equations are inconsistent, i.e.  $(\mathbf{C} - \lambda \mathbf{I}) \mathbf{u} \neq \mathbf{b}$ , or if  $\mathbf{u}^T \mathbf{u} > s^2$ , then there is no solution for this  $\lambda$ . On the other hand, if the equations are consistent, we have a unique solution  $\mathbf{z} = \mathbf{u}$  for  $\mathbf{u}^T \mathbf{u} = s^2$ , and several solutions for  $\mathbf{u}^T \mathbf{u} < s^2$ .

**PROOF.** If  $(\mathbf{C} - \lambda \mathbf{I}) \mathbf{u} = (\mathbf{C} - \lambda \mathbf{I})(\mathbf{C} - \lambda \mathbf{I})^+ \mathbf{b} \neq \mathbf{b}$ , then  $\mathbf{b} \notin \mathcal{R}(\mathbf{C} - \lambda \mathbf{I})$ , and the Lagrange equations have no solution for this  $\lambda$ . Let be  $\mathbf{b} \in \mathcal{R}(\mathbf{C} - \lambda \mathbf{I})$  in the following. Then  $(\mathbf{C} - \lambda \mathbf{I}) \mathbf{u} = (\mathbf{C} - \lambda \mathbf{I})(\mathbf{C} - \lambda \mathbf{I})^+ \mathbf{b} = \mathbf{b}$ , and the first Lagrange equation is satisfied.

The vector  $\mathbf{u}$  denotes the solution with smallest norm of the equation  $\mathbf{C} \mathbf{u} = \lambda \mathbf{u} + \mathbf{b}$ . Therefore, if  $\|\mathbf{u}\| > s$ , the normalization constraint  $\mathbf{z}^T \mathbf{z} = s^2$  cannot be satisfied, and we have no solution of the Lagrange equations for this  $\lambda$ . In the case of  $\|\mathbf{u}\| = s$ , the vector  $\mathbf{u}$  denotes the unique solution.

Finally, let us assume that  $\|\mathbf{u}\| < s$ . The vectors  $\xi^{(1)}, \dots, \xi^{(k)}$  shall be the  $k$  orthonormal eigenvectors corresponding to the

eigenvalue  $\lambda$ . As

$$\begin{aligned}\mathbf{u} &\perp \mathcal{N}(C - \lambda I) \\ \xi^{(i)} &\in \mathcal{N}(C - \lambda I),\end{aligned}$$

all the vectors

$$\mathbf{z} = \mathbf{u} + c_1 \xi^{(1)} + \cdots + c_k \xi^{(k)}$$

with

$$c_1^2 + \cdots + c_k^2 = s^2 - \|\mathbf{u}\|^2$$

are solutions of the Lagrange equations. Therefore, the set of solutions constitutes a manifold of dimension  $k - 1$ .  $\square$

Discussing the solvability of the Lagrange equations (13) we derived the result that the solution with the smallest  $\lambda$  must satisfy  $\lambda \leq \delta_1$ , where  $\delta_1$  denotes the smallest eigenvalue of  $C$ . Let  $\lambda$  be hereafter the smallest eigenvalue of the quadratic eigenvalue equation (32). Together with the forementioned theorems we can make the following distinction:

1. It can hold that  $\lambda < \delta_1$ . Then  $\lambda$  lies outside the spectrum  $\lambda(C)$ , and  $\mathbf{z} := (C - \lambda I)^{-1} \mathbf{b}$  fulfils the Lagrange equations (13). The solution is unique.
2. Now let  $\lambda = \delta_1$ . The vector  $\mathbf{u} := (C - \lambda I)^+ \mathbf{b}$  must satisfy the equation  $C\mathbf{u} = \lambda \mathbf{u} + \mathbf{b}$ . If  $\mathbf{u}^T \mathbf{u} = s^2$ , then  $\mathbf{z} := \mathbf{u}$  is the unique solution of the Lagrange equations (13).
3. Finally, if  $\mathbf{u}^T \mathbf{u} < s^2$ , we must find an eigenvector  $\xi$  to the eigenvalue  $\lambda$  of  $C$  with  $\xi^T \xi = s^2 - \mathbf{u}^T \mathbf{u}$ . Then,  $\mathbf{z} := \mathbf{u} + \xi$  represents one of the many solutions of the Lagrange equations (13).

**5.2 Solving the Quadratic Eigenvalue Problem** The quadratic eigenvalue problem (32) can be reduced to an ordinary eigenvalue problem by properly chosen transformations. With the definition

$$(33) \quad \boldsymbol{\eta} := (C - \lambda I) \boldsymbol{\gamma}$$

the following equations can be established:

$$\begin{aligned} C\gamma - \eta &= \lambda\gamma \\ C\eta - \frac{1}{s^2}\mathbf{b}\mathbf{b}^T\gamma &= \lambda\eta. \end{aligned}$$

In matrix terms this leads to:

$$(34) \quad \begin{bmatrix} C & -I \\ -\frac{1}{s^2}\mathbf{b}\mathbf{b}^T & C \end{bmatrix} \begin{bmatrix} \gamma \\ \eta \end{bmatrix} = \lambda \begin{bmatrix} \gamma \\ \eta \end{bmatrix}.$$

Thus we have transformed the original quadratic eigenvalue problem into an equivalent linear one, that can be solved with traditional methods.

**6 NUMERICAL RESULTS** Finally we present some results of the numerical experiments. All calculations were carried out on a VAX 8600 with Floating Point Accelerator (single precision, 32 bit reals) under VMS 4.6. The IMSL-Library Edition 10.0 / Version 1.0 served for the basic computations. The input data was produced by a random number generator. As a reference solution we employed an implementation in double precision. The tables 1 to 3 summarize the results. The individual columns have the following meaning:

$\lambda$  : smallest zero of the secular equation  
 min : minimum  
 Norm: norm of the solution  $\mathbf{x}$   
 Error: norm of the difference  $\mathbf{x} - \mathbf{x}_{\text{Reference Solution}}$   
 Time: CPU-time

Furthermore the caption contains the dimension  $n$  of the problem together with the condition numbers (24, 25) of the solutions.



Table 1:  $n = 10$   $\|\kappa(\mathbf{x})\| = 0.104$   $\kappa(\min) = -0.282$ 

| Algorithm      | $\lambda$ | min       | Norm     | Error               | Time     |
|----------------|-----------|-----------|----------|---------------------|----------|
| Reference      | -4.220885 | -0.677439 | 1.000000 |                     |          |
| Expl. Sec. Eq. | -4.220886 | -0.677439 | 1.000000 | $2.0 \cdot 10^{-7}$ | 0.05 sec |
| Impl. Sec. Eq. | -4.220886 | -0.677439 | 1.000000 | $6.5 \cdot 10^{-8}$ | 0.10 sec |
| Quadr. EVP     | -4.220890 | -0.677438 | 1.000000 | $5.4 \cdot 10^{-7}$ | 0.23 sec |

Table 2:  $n = 45$   $\|\kappa(\mathbf{x})\| = 930.8$   $\kappa(\min) = -13131$ 

| Algorithm      | $\lambda$ | min       | Norm     | Error               | Time     |
|----------------|-----------|-----------|----------|---------------------|----------|
| Reference      | -7.054650 | -7.054494 | 1.000000 |                     |          |
| Expl. Sec. Eq. | -7.054657 | -7.052025 | 0.999825 | $1.8 \cdot 10^{-4}$ | 1.16 sec |
| Impl. Sec. Eq. | -7.054650 | -7.060933 | 1.000456 | $4.6 \cdot 10^{-4}$ | 2.60 sec |
| Quadr. EVP     | -7.055462 | -2.290894 | 0.569900 | $4.3 \cdot 10^{-1}$ | 10.4 sec |

Based on the numerical calculations the three solutions can be judged as follows:

**Explicit secular equation.** The zero of the secular equation (20) is determined to machine precision. The accuracy that can be expected by the condition numbers (24, 25) is achieved.

**Implicit secular equation.** This method achieves the same accuracy as the first one. However the calculation of an eigenvalue decomposition (14) is replaced by the determination of a generalized inverse which is in no way cheaper than the former operation.

**Quadratic eigenvalue problem.** It turns out that the smallest eigenvalue of the general matrix (34) can be calculated only very inexactly. We suppose that the transformation into a quadratic eigenvalue problem (32) impairs the condition of the problem. With large condition numbers (24, 25) all decimal places can be incorrect.

Table 3:  $n = 100$   $\|\kappa(\mathbf{x})\| = 6.019$   $\kappa(\min) = -125.2$ 

| Algorithm      | $\lambda$ | min       | Norm     | Error               | Time     |
|----------------|-----------|-----------|----------|---------------------|----------|
| Reference      | -11.48939 | -11.07225 | 1.000000 |                     |          |
| Expl. Sec. Eq. | -11.48939 | -11.07225 | 1.000000 | $5.3 \cdot 10^{-6}$ | 9.90 sec |
| Impl. Sec. Eq. | -11.48939 | -11.07230 | 1.000002 | $2.2 \cdot 10^{-6}$ | 24.5 sec |
| Quadr. EVP     | -11.48976 | -11.02595 | 0.997983 | $2.2 \cdot 10^{-3}$ | 104. sec |

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